

Boundary-Constrained Rare-Fault Triggering for Short-Horizon Hoister Fault-State Prediction

Abstract

Industrial hoisting monitors must recognize rare degradation states before they are hidden by dominant normal and stop states. We study fixed short-horizon future-state prediction on a private five-state Hoister overspeed dataset where the second-level degradation state appears in only 185 of 37,417 timestamps. The tested time-series classifiers can achieve high overall accuracy while collapsing to zero F1 on this rare state. SGTONetV6 separates conservative multiclass future-state prediction from a calibrated rare-fault trigger constrained by boundary and precursor semantics. Across three file-level splits with `label_shift=1`, SGTONetV6 obtains macro-F1 0.6233, balanced accuracy 0.6731, fault macro-F1 0.5411, and class-9 F1 0.5556. The current evidence is an oracle boundary-gated diagnostic evaluation, not a complete online deployment protocol. The evidence supports scoped short-horizon rare-boundary recovery; `label_shift=3` remains a limitation.

Keywords— Hoisting systems, future-state prediction, rare-event detection, time-series classification, class imbalance.

1 Introduction

Industrial hoisting systems operate through ordered states that can include normal running, early degradation, stronger degradation, and fault occurrence. For monitoring, retrospective state recognition is not enough: a controller or operator benefits from predicting the next short-horizon state from the current sensor window. This digest studies that fixed-horizon future-state classification problem for a Hoister overspeed process.

The central difficulty is rare-boundary collapse. In the available 27-file private Hoister dataset, the five raw labels are 1, 5, 7, 9, and 3, with label 9 representing the rare second-level degradation state. Only 185 of 37,417 timestamps belong to class 9, less than 0.5 percent of the sequence. Under such imbalance, a model can appear useful by aggregate accuracy while never predicting the safety-relevant rare state. In our `label_shift=1` experiments, iTransformer reaches the highest accuracy, 0.8175, but its class-9 F1 is 0.0000. DLinear, TimesNet, PatchTST, and a conservative SGTONetV4 baseline also have class-9 F1 of 0.0000.

We propose SGTONetV6, a shift-aware graph and trigger-oriented network for this short-horizon setting. The design intentionally separates two behaviors. A conservative future-state classifier models the five-way state transition, while a local rare-fault trigger can override the prediction to class 9 only when the rare score is high and the sample satisfies boundary and precursor constraints. This makes rare recovery a constrained operational decision instead of a global reweighting of the multiclass classifier.

The contribution is therefore scoped. SGTONetV6 is not presented as a generally higher-accuracy classifier. Rather, it addresses a specific failure mode: high-accuracy temporal backbones can miss a rare transition state entirely. On `label_shift=1`, SGTONetV6 recovers class 9 with F1 0.5556 and improves fault macro-F1 to 0.5411, while all tested non-trigger baselines have zero class-9 F1. The present results use label-derived boundary flags for the override gate, so they should be read as oracle boundary-gated diagnostic evidence. At `label_shift=3`, the current trigger loses precision; we report that result as a limitation rather than a positive multi-horizon claim.

2 Task and Method

Let the input be a multivariate sensor window ending at time t . The task is to predict the label one step ahead; the main evidence in this digest uses `label_shift 1`, sequence length 96, and window step 8. The data loader constructs each sample with a future label, current label, and boundary flag when future-state targets are enabled. The direct overspeed indicator `JianSuDuan_ChaoSu` is excluded from the input columns to avoid leaking fault-state information.

SGTONetV6 inherits the SGTO patch temporal encoder and uses a horizon embedding, a boundary head, graph-aware future-state refinement, destination experts, prototype logits, and a future-state head to produce the conservative five-class prediction. For rare recovery, a learned rare query attends over patch tokens to form a localized rare context. The rare trigger head combines the hidden state, future hidden state, rare context, current and future probability vectors, rare prototype similarity, and boundary logit into a scalar rare score.

At evaluation time, the rare trigger is constrained by the semantics of the transition. The prediction is changed to class 9 only if the rare score exceeds the calibrated threshold, the sample is marked as a transition boundary, and the current label is a plausible precursor in $\{5, 7\}$. Otherwise the conservative classifier prediction is kept. In the current implementation, the boundary flag is derived from current and future labels and the precursor gate uses the current label supplied by the dataset. Thus the reported override result is an oracle boundary-gated diagnostic protocol. A deployable online version must replace these gates with a predicted boundary signal and an observable or estimated current-state signal.

Validation-based threshold calibration is used when possible; because class 9 is extremely sparse, the implementation also permits a fallback threshold prior when validation rare samples are insufficient.

This rule is deliberately simple. It encodes the assumption that second-level degradation should be triggered near plausible state transitions, not everywhere a binary classifier is uncertain. The ablation results below test whether this boundary-constrained trigger, rather than backbone capacity alone, is responsible for the recovered rare-state behavior.

3 Experimental Protocol

The dataset contains 27 CSV files, 37,417 timestamps, and 20 columns. The target is `running_state_five_class`. The five-class distribution is: label 1, 10,959; label 5, 15,695; label 7, 5,364; label

Table 1: Main label_shift=1 results. Mean values over three file-level splits.

Model	Acc.	Macro-F1	Bal. Acc.	Fault Macro-F1	Class-9 F1
SGTONetV6	0.7102	0.6233	0.6731	0.5411	0.5556
iTransformer	0.8175	0.6185	0.6594	0.4615	0.0000
DLinear	0.7895	0.5961	0.6466	0.4426	0.0000
TimesNet	0.7958	0.5893	0.6207	0.4275	0.0000
PatchTST	0.7559	0.5687	0.6154	0.4194	0.0000
SGTONetV4	0.7630	0.5605	0.5961	0.3999	0.0000

Table 2: Mechanism ablation under label_shift=1.

Variant	Macro-F1	Bal. Acc.	Class-9 F1	Interpretation
Full SGTONetV6	0.6233	0.6731	0.5556	Full constrained trigger
No precursor constraint	0.6139	0.6725	0.5101	Precursor prior helps
Mean rare context	0.5848	0.6654	0.2317	Patch attention matters
No fallback prior	0.5830	0.6397	0.3556	Fallback helps sparse validation
No rare override	0.5113	0.5568	0.0000	Trigger is necessary
No boundary constraint	0.4550	0.5469	0.0158	Boundary constraint is critical

9, 185; and label 3, 5,214. All reported main results use file-level split seeds 14, 22, and 30, batch size 16, class weights, and macro-F1-oriented early stopping.

Baselines are DLinear, TimesNet, iTransformer, PatchTST, and SGTONetV4Conservative. Metrics include accuracy, macro-F1, balanced accuracy, fault macro-F1, and class-9 precision, recall, and F1. Accuracy is reported for context but is not the primary criterion because it is dominated by common states.

4 Results and Discussion

Table 1 shows the main short-horizon result. SGTONetV6 has lower overall accuracy than iTransformer, DLinear, TimesNet, PatchTST, and SGTONetV4, so the result should not be read as an accuracy improvement. It has the numerically highest macro-F1, balanced accuracy, and fault macro-F1 in this three-split study, but the robust effect is that it is the only tested method that recovers class 9.

The small macro-F1 margin over iTransformer, 0.6233 versus 0.6185, should be interpreted conservatively. The corresponding split standard deviations are 0.0332 and 0.0199, respectively. The stronger evidence is the change in rare-state behavior: all non-trigger baselines report class-9 precision, recall, and F1 of 0.0000, while SGTONetV6 reaches class-9 precision 0.5611, recall 0.5833, and F1 0.5556 with class-9 F1 standard deviation 0.1133.

Table 2 supports the proposed mechanism within the oracle boundary-gated evaluation protocol. Removing the rare override collapses class-9 F1 to 0.0000, indicating that the conservative future-state classifier alone does not solve the rare-boundary problem. Removing the boundary constraint drops class-9 F1 to 0.0158 and macro-F1 to 0.4550, showing that an unconstrained trigger causes uncontrolled rare predictions. Replacing patch-attentive rare context with mean context reduces class-9 F1 to 0.2317, suggesting that rare evidence is localized within the window. Removing the fallback threshold prior also reduces class-9 F1, consistent with the difficulty of calibrating on validation splits that may contain very few rare

Table 3: Horizon limitation. The current method is not a general multi-horizon solution.

Horizon	Model	Macro-F1	Bal. Acc.	C9 Prec.	C9 Rec.	C9 F1
1	SGTONetV6	0.6233	0.6731	0.5611	0.5833	0.5556
1	PatchTST	0.5687	0.6154	0.0000	0.0000	0.0000
3	PatchTST	0.5877	0.6310	0.1789	0.2597	0.1919
3	SGTONetV6	0.5006	0.6112	0.0620	0.4762	0.1070
3	SGTONetV4	0.4805	0.5466	0.0000	0.0000	0.0000

samples.

The threshold-sensitivity curve in the saved results gives the same interpretation. The best tested global threshold is near 0.009, with macro-F1 0.6069 and class-9 F1 0.4732. This is lower than the three-split calibrated main result, so the curve is used as calibration evidence rather than as the headline performance number.

Table 3 bounds the claim. At label_shift=3, PatchTST reaches macro-F1 0.5877 and class-9 F1 0.1919, while SGTONetV6 reaches macro-F1 0.5006 and class-9 F1 0.1070. SGTONetV6 keeps higher rare recall than precision at this horizon, which indicates many false positives. Therefore the present evidence supports short-horizon rare-boundary recovery at label_shift=1 under the current diagnostic gating protocol, not a broad multi-horizon or fully online superiority claim.

5 Conclusion

This digest identifies rare-boundary collapse in short-horizon Hoister future-state prediction: the tested high-accuracy temporal classifiers can miss the rare second-level degradation state entirely. SGTONetV6 addresses this failure mode by separating conservative future-state classification from a calibrated rare-fault trigger constrained by boundary and precursor semantics. On the available private Hoister dataset with label_shift=1, the method recovers class 9 and improves fault-oriented macro metrics under an oracle boundary-gated diagnostic protocol, while preserving a clear limitation: the same calibration does not transfer cleanly to label_shift=3.

The remaining risks are also clear. The evidence comes from one private dataset, the accuracy tradeoff is real, the macro-F1 gain over iTransformer is small, rare-trigger calibration depends on sparse validation evidence, and the current boundary gate is not yet an online inference signal. Future work should evaluate multi-site or public data, tune horizon-specific trigger calibration, and quantify deployment false-alarm costs.

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